6. Suppose that in a certain drug the concentration of a particular chemical is a random variable with a continuous distribution for which the p.d.f. g is as follows:

$$g(x) = \begin{cases} \frac{3}{8}x^2 & \text{for } 0 \le x \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that the concentrations X and Y of the chemical in two separate batches of the drug are independent random variables for each of which the p.d.f. is g. Determine **(a)** the joint p.d.f. of X and Y; **(b)** $\Pr(X = Y)$; **(c)** $\Pr(X > Y)$; **(d)** $\Pr(X + Y \le 1)$.

7. Suppose that the joint p.d.f. of X and Y is as follows:

$$f(x, y) = \begin{cases} 2xe^{-y} & \text{for } 0 \le x \le 1 \text{ and } 0 < y < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

Are X and Y independent?

8. Suppose that the joint p.d.f. of X and Y is as follows:

$$f(x, y) = \begin{cases} 24xy & \text{for } x \ge 0,\ y \ge 0, \text{ and } x + y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Are X and Y independent?

9. Suppose that a point (X, Y) is chosen at random from the rectangle S defined as follows:

$$S = \{(x, y) : 0 \le x \le 2 \text{ and } 1 \le y \le 4\}.$$

a. Determine the joint p.d.f. of X and Y, the marginal p.d.f. of X, and the marginal p.d.f. of Y.

b. Are X and Y independent?

10. Suppose that a point (X, Y) is chosen at random from the circle S defined as follows:

$$S = \{(x, y) : x^2 + y^2 \le 1\}.$$

a. Determine the joint p.d.f. of X and Y, the marginal p.d.f. of X, and the marginal p.d.f. of Y.

b. Are X and Y independent?

11. Suppose that two persons make an appointment to meet between 5 P.M. and 6 P.M. at a certain location, and they agree that neither person will wait more than 10 minutes for the other person. If they arrive independently at random times between 5 P.M. and 6 P.M., what is the probability that they will meet?

12. Prove Theorem 3.5.6.

13. In Example 3.5.10, verify that X and Y have the same marginal p.d.f.'s and that

$$f_1(x) = \begin{cases} 2kx^2(1 - x^2)^{3/2}/3 & \text{if } -1 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

14. For the joint p.d.f. in Example 3.4.7, determine whether or not X and Y are independent.

15. A painting process consists of two stages. In the first stage, the paint is applied, and in the second stage, a protective coat is added. Let X be the time spent on the first stage, and let Y be the time spent on the second stage. The first stage involves an inspection. If the paint fails the inspection, one must wait three minutes and apply the paint again. After a second application, there is no further inspection. The joint p.d.f. of X and Y is

$$f(x, y) = \begin{cases} \frac{1}{3} & \text{if } 1 < x < 3 \text{ and } 0 < y < 1, \\ \frac{1}{6} & \text{if } 6 < x < 8 \text{ and } 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

a. Sketch the region where $f(x, y) > 0$. Note that it is not exactly a rectangle.

b. Find the marginal p.d.f.'s of X and Y.

c. Show that X and Y are independent.

This problem does not contradict Theorem 3.5.6. In that theorem the conditions, including that the set where $f(x, y) > 0$ be rectangular, are sufficient but not necessary.

3.6 Conditional Distributions

We generalize the concept of conditional probability to conditional distributions. Recall that distributions are just collections of probabilities of events determined by random variables. Conditional distributions will be the probabilities of events determined by some random variables conditional on events determined by other random variables. The idea is that there will typically be many random variables of interest in an applied problem. After we observe some of those random variables, we want to be able to adjust the probabilities associated with the ones that have not yet been observed. The conditional distribution of one random variable X given another Y will be the distribution that we would use for X after we learn the value of Y.

Table 3.7 Joint p.f. for Example 3.6.1

	Brand Y					
Stolen X	1	2	3	4	5	Total
0	0.129	0.298	0.161	0.280	0.108	0.976
1	0.010	0.010	0.001	0.002	0.001	0.024
Total	0.139	0.308	0.162	0.282	0.109	1.000

Discrete Conditional Distributions

Example 3.6.1

Auto Insurance. Insurance companies keep track of how likely various cars are to be stolen. Suppose that a company in a particular area computes the joint distribution of car brands and the indicator of whether the car will be stolen during a particular year that appears in Table 3.7.

We let $X = 1$ mean that a car is stolen, and we let $X = 0$ mean that the car is not stolen. We let Y take one of the values from 1 to 5 to indicate the brand of car as indicated in Table 3.7. If a customer applies for insurance for a particular brand of car, the company needs to compute the distribution of the random variable X as part of its premium determination. The insurance company might adjust their premium according to a risk factor such as likelihood of being stolen. Although, overall, the probability that a car will be stolen is 0.024, if we assume that we know the brand of car, the probability might change quite a bit. This section introduces the formal concepts for addressing this type of problem. ◀

Suppose that X and Y are two random variables having a discrete joint distribution for which the joint p.f. is f. As before, we shall let f_1 and f_2 denote the marginal p.f.'s of X and Y, respectively. After we observe that $Y = y$, the probability that the random variable X will take a particular value x is specified by the following conditional probability:

$$\Pr(X = x|Y = y) = \frac{\Pr(X = x \text{ and } Y = y)}{\Pr(Y = y)} = \frac{f(x, y)}{f_2(y)}. \tag{3.6.1}$$

In other words, if it is known that $Y = y$, then the probability that $X = x$ will be updated to the value in Eq. (3.6.1). Next, we consider the entire distribution of X after learning that $Y = y$.

Definition 3.6.1

Conditional Distribution/p.f. Let X and Y have a discrete joint distribution with joint p.f. f. Let f_2 denote the marginal p.f. of Y. For each y such that $f_2(y) > 0$, define

$$g_1(x|y) = \frac{f(x, y)}{f_2(y)}. \tag{3.6.2}$$

Then g_1 is called the *conditional p.f. of X given Y*. The discrete distribution whose p.f. is $g_1(\cdot|y)$ is called the *conditional distribution of X given that $Y = y$*.

Table 3.8 Conditional p.f. of Y given X for Example 3.6.3

	Brand Y				
Stolen X	1	2	3	4	5
0	0.928	0.968	0.994	0.993	0.991
1	0.072	0.032	0.006	0.007	0.009

We should verify that $g_1(x|y)$ is actually a p.f. as a function of x for each y. Let y be such that $f_2(y) > 0$. Then $g_1(x|y) \geq 0$ for all x and

$$\sum_x g_1(x|y) = \frac{1}{f_2(y)} \sum_x f(x, y) = \frac{1}{f_2(y)} f_2(y) = 1.$$

Notice that we do not bother to define $g_1(x|y)$ for those y such that $f_2(y) = 0$.

Similarly, if x is a given value of X such that $f_1(x) = \Pr(X = x) > 0$, and if $g_2(y|x)$ is the *conditional p.f. of Y given that $X = x$*, then

$$g_2(y|x) = \frac{f(x, y)}{f_1(x)}. \tag{3.6.3}$$

For each x such that $f_1(x) > 0$, the function $g_2(y|x)$ will be a p.f. as a function of y.

Example 3.6.2 Calculating a Conditional p.f. from a Joint p.f. Suppose that the joint p.f. of X and Y is as specified in Table 3.4 in Example 3.5.2. We shall determine the conditional p.f. of Y given that $X = 2$.

The marginal p.f. of X appears in the Total column of Table 3.4, so $f_1(2) = \Pr(X = 2) = 0.6$. Therefore, the conditional probability $g_2(y|2)$ that Y will take a particular value y is

$$g_2(y|2) = \frac{f(2, y)}{0.6}.$$

It should be noted that for all possible values of y, the conditional probabilities $g_2(y|2)$ must be proportional to the joint probabilities $f(2, y)$. In this example, each value of $f(2, y)$ is simply divided by the constant $f_1(2) = 0.6$ in order that the sum of the results will be equal to 1. Thus,

$$g_2(1|2) = 1/2, \quad g_2(2|2) = 0, \quad g_2(3|2) = 1/6, \quad g_2(4|2) = 1/3. \quad \blacktriangleleft$$

Example 3.6.3 Auto Insurance. Consider again the probabilities of car brands and cars being stolen in Example 3.6.1. The conditional distribution of X (being stolen) given Y (brand) is given in Table 3.8. It appears that Brand 1 is much more likely to be stolen than other cars in this area, with Brand 1 also having a significant chance of being stolen. $\blacktriangleleft$

Continuous Conditional Distributions

Example 3.6.4 Processing Times. A manufacturing process consists of two stages. The first stage takes Y minutes, and the whole process takes X minutes (which includes the first

Y minutes). Suppose that X and Y have a joint continuous distribution with joint p.d.f.

$$f(x, y) = \begin{cases} e^{-x} & \text{for } 0 \le y \le x < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

After we learn how much time Y that the first stage takes, we want to update our distribution for the total time X. In other words, we would like to be able to compute a conditional distribution for X given $Y = y$. We cannot argue the same way as we did with discrete joint distributions, because $\{Y = y\}$ is an event with probability 0 for all y. ◀

To facilitate the solutions of problems such as the one posed in Example 3.6.4, the concept of conditional probability will be extended by considering the definition of the conditional p.f. of X given in Eq. (3.6.2) and the analogy between a p.f. and a p.d.f.

Definition 3.6.2 Conditional p.d.f. Let X and Y have a continuous joint distribution with joint p.d.f. f and respective marginals f_1 and f_2. Let y be a value such that $f_2(y) > 0$. Then the *conditional p.d.f. g_1 of X given that $Y = y$* is defined as follows:

$$g_1(x|y) = \frac{f(x, y)}{f_2(y)} \quad \text{for } -\infty < x < \infty. \tag{3.6.4}$$

For values of y such that $f_2(y) = 0$, we are free to define $g_1(x|y)$ however we wish, so long as $g_1(x|y)$ is a p.d.f. as a function of x.

It should be noted that Eq. (3.6.2) and Eq. (3.6.4) are identical. However, Eq. (3.6.2) was *derived* as the conditional probability that $X = x$ given that $Y = y$, whereas Eq. (3.6.4) was *defined* to be the value of the conditional p.d.f. of X given that $Y = y$. In fact, we should verify that $g_1(x|y)$ as defined above really is a p.d.f.

Theorem 3.6.1 For each y, $g_1(x|y)$ defined in Definition 3.6.2 is a p.d.f. as a function of x.

Proof If $f_2(y) = 0$, then g_1 is defined to be any p.d.f. we wish, and hence it is a p.d.f. If $f_2(y) > 0$, g_1 is defined by Eq. (3.6.4). For each such y, it is clear that $g_1(x|y) \ge 0$ for all x. Also, if $f_2(y) > 0$, then

$$\int_{-\infty}^{\infty} g_1(x|y)\, dx = \frac{\int_{-\infty}^{\infty} f(x, y)\, dx}{f_2(y)} = \frac{f_2(y)}{f_2(y)} = 1,$$

by using the formula for $f_2(y)$ in Eq. (3.5.3). ■

Example 3.6.5 Processing Times. In Example 3.6.4, Y is the time that the first stage of a process takes, while X is the total time of the two stages. We want to calculate the conditional p.d.f. of X given Y. We can calculate the marginal p.d.f. of Y as follows: For each y, the possible values of X are all $x \ge y$, so for each $y > 0$,

$$f_2(y) = \int_y^{\infty} e^{-x} dx = e^{-y},$$

and $f_2(y) = 0$ for $y < 0$. For each $y \ge 0$, the conditional p.d.f. of X given $Y = y$ is then

$$g_1(x|y) = \frac{f(x, y)}{f_2(y)} = \frac{e^{-x}}{e^{-y}} = e^{y-x}, \text{ for } x \ge y,$$

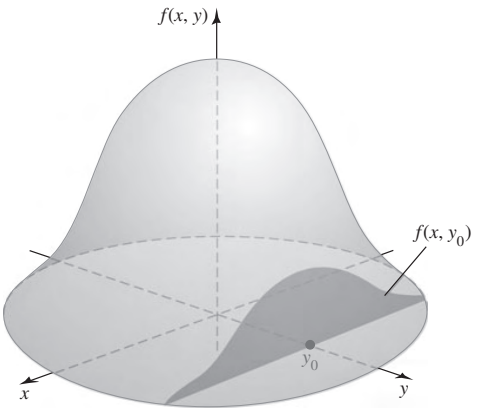


Figure 3.20 The conditional p.d.f. $g_1(x|y_0)$ is proportional to $f(x, y_0)$.

and $g_1(x|y) = 0$ for $x < y$. So, for example, if we observe $Y = 4$ and we want the conditional probability that $X \geq 9$, we compute

$$\Pr(X \geq 9|Y = 4) = \int_9^\infty e^{4-x}dx = e^{-5} = 0.0067. \quad \blacktriangleleft$$

Definition 3.6.2 has an interpretation that can be understood by considering Fig. 3.20. The joint p.d.f. f defines a surface over the xy-plane for which the height $f(x, y)$ at each point (x, y) represents the relative likelihood of that point. For instance, if it is known that $Y = y_0$, then the point (x, y) must lie on the line $y = y_0$ in the xy-plane, and the relative likelihood of any point (x, y_0) on this line is $f(x, y_0)$. Hence, the conditional p.d.f. $g_1(x|y_0)$ of X should be proportional to $f(x, y_0)$. In other words, $g_1(x|y_0)$ is essentially the same as $f(x, y_0)$, but it includes a constant factor $1/[f_2(y_0)]$, which is required to make the conditional p.d.f. integrate to unity over all values of x.

Similarly, for each value of x such that $f_1(x) > 0$, the *conditional p.d.f. of Y given that $X = x$* is defined as follows:

$$g_2(y|x) = \frac{f(x, y)}{f_1(x)} \quad \text{for } -\infty < y < \infty. \tag{3.6.5}$$

This equation is identical to Eq. (3.6.3), which was derived for discrete distributions. If $f_1(x) = 0$, then $g_2(y|x)$ is arbitrary so long as it is a p.d.f. as a function of y.

Example 3.6.6

Calculating a Conditional p.d.f. from a Joint p.d.f. Suppose that the joint p.d.f. of X and Y is as specified in Example 3.4.8 on page 122. We shall first determine the conditional p.d.f. of Y given that $X = x$ and then determine some probabilities for Y given the specific value $X = 1/2$.

The set S for which $f(x, y) > 0$ was sketched in Fig. 3.12 on page 123. Furthermore, the marginal p.d.f. f_1 was derived in Example 3.5.3 on page 132 and sketched in Fig. 3.17 on page 133. It can be seen from Fig. 3.17 that $f_1(x) > 0$ for $-1 < x < 1$ but not for $x = 0$. Therefore, for each given value of x such that $-1 < x < 0$ or $0 < x < 1$, the conditional p.d.f. $g_2(y|x)$ of Y will be as follows:

$$g_2(y|x) = \begin{cases} \dfrac{2y}{1 - x^4} & \text{for } x^2 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, if it is known that $X = 1/2$, then $\Pr\left(Y \geq \frac{1}{4} \middle| X = \frac{1}{2}\right) = 1$ and

$$\Pr\left(Y \geq \frac{3}{4} \middle| X = \frac{1}{2}\right) = \int_{3/4}^{1} g_2\left(y \middle| \frac{1}{2}\right) dy = \frac{7}{15}. \quad \blacktriangleleft$$

Note: A Conditional p.d.f. Is Not the Result of Conditioning on a Set of Probability Zero. The conditional p.d.f. $g_1(x|y)$ of X given $Y = y$ is the p.d.f. we would use for X if we were to learn that $Y = y$. This sounds as if we were conditioning on the event $\{Y = y\}$, which has zero probability if Y has a continuous distribution. Actually, for the cases we shall see in this text, the value of $g_1(x|y)$ is a limit:

$$g_1(x|y) = \lim_{\epsilon \to 0} \frac{\partial}{\partial x} \Pr(X \leq x | y - \epsilon < Y \leq y + \epsilon). \tag{3.6.6}$$

The conditioning event $\{y - \epsilon \leq Y \leq y + \epsilon\}$ in Eq. (3.6.6) has positive probability if the marginal p.d.f. of Y is positive at y. The mathematics required to make this rigorous is beyond the scope of this text. (See Exercise 11 in this section and Exercises 25 and 26 in Sec. 3.11 for results that we can prove.) Another way to think about conditioning on a continuous random variable is to notice that the conditional p.d.f.'s that we compute are typically continuous as a function of the conditioning variable. This means that conditioning on $Y = y$ or on $Y = y + \epsilon$ for small ϵ will produce nearly the same conditional distribution for X. So it does not matter much if we use $Y = y$ as a surogate for Y close to y. Nevertheless, it is important to keep in mind that the conditional p.d.f. of X given $Y = y$ is better thought of as the conditional p.d.f. of X given that Y is very close to y. This wording is awkward, so we shall not use it, but we must remember the distinction between the conditional p.d.f. and conditioning on an event with probability 0. Despite this distinction, it is still legitimate to treat Y as the constant y when dealing with the conditional distribution of X given $Y = y$.

For mixed joint distributions, we continue to use Eqs. (3.6.2) and (3.6.3) to define conditional p.f.'s and p.d.f.'s.

Definition 3.6.3 Conditional p.f. or p.d.f. from Mixed Distribution. Let X be discrete and let Y be continuous with joint p.f./p.d.f. f. Then the *conditional p.f. of X given $Y = y$* is defined by Eq. (3.6.2), and the *conditional p.d.f. of Y given $X = x$* is defined by Eq. (3.6.3).

Construction of the Joint Distribution

Example 3.6.7 Defective Parts. Suppose that a certain machine produces defective and nondefective parts, but we do not know what proportion of defectives we would find among all parts that could be produced by this machine. Let P stand for the unknown proportion of defective parts among all possible parts produced by the machine. If we were to learn that $P = p$, we might be willing to say that the parts were independent of each other and each had probability p of being defective. In other words, if we condition on $P = p$, then we have the situation described in Example 3.1.9. As in that example, suppose that we examine n parts and let X stand for the number of defectives among the n examined parts. The distribution of X, assuming that we know $P = p$, is the binomial distribution with parameters n and p. That is, we can let the binomial p.f. (3.1.4) be the conditional p.f. of X given $P = p$, namely,

$$g_1(x|p) = \binom{n}{x} p^x (1-p)^{n-x}, \text{ for } x = 0, \ldots, n.$$

We might also believe that P has a continuous distribution with p.d.f. such as $f_2(p) = 1$ for $0 \le p \le 1$. (This means that P has the uniform distribution on the interval $[0, 1]$.) We know that the conditional p.f. g_1 of X given $P = p$ satisfies

$$g_1(x|p) = \frac{f(x, p)}{f_2(p)},$$

where f is the joint p.f./p.d.f. of X and P. If we multiply both sides of this equation by $f_2(p)$, it follows that the joint p.f./p.d.f. of X and P is

$$f(x, p) = g_1(x|p) f_2(p) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \text{for } x = 0, \ldots, n, \text{ and } 0 \le p \le 1.$$

◀

The construction in Example 3.6.7 is available in general, as we explain next.

Generalizing the Multiplication Rule for Conditional Probabilities A special case of Theorem 2.1.2, the multiplication rule for conditional probabilities, says that if A and B are two events, then $\Pr(A \cap B) = \Pr(A) \Pr(B|A)$. The following theorem, whose proof is immediate from Eqs. (3.6.4) and (3.6.5), generalizes Theorem 2.1.2 to the case of two random variables.

Theorem 3.6.2 Multiplication Rule for Distributions. Let X and Y be random variables such that X has p.f. or p.d.f. $f_1(x)$ and Y has p.f. or p.d.f. $f_2(y)$. Also, assume that the conditional p.f. or p.d.f. of X given $Y = y$ is $g_1(x|y)$ while the conditional p.f. or p.d.f. of Y given $X = x$ is $g_2(y|x)$. Then for each y such that $f_2(y) > 0$ and each x,

$$f(x, y) = g_1(x|y) f_2(y), \tag{3.6.7}$$

where f is the joint p.f., p.d.f., or p.f./p.d.f. of X and Y. Similarly, for each x such that $f_1(x) > 0$ and each y,

$$f(x, y) = f_1(x) g_2(y|x). \tag{3.6.8}$$

■

In Theorem 3.6.2, if $f_2(y_0) = 0$ for some value y_0, then it can be assumed without loss of generality that $f(x, y_0) = 0$ for all values of x. In this case, both sides of Eq. (3.6.7) will be 0, and the fact that $g_1(x|y_0)$ is not uniquely defined becomes irrelevant. Hence, Eq. (3.6.7) will be satisfied for *all* values of x and y. A similar statement applies to Eq. (3.6.8).

Example 3.6.8 Waiting in a Queue. Let X be the amount of time that a person has to wait for service in a queue. The faster the server works in the queue, the shorter should be the waiting time. Let Y stand for the rate at which the server works, which we will take to be unknown. A common choice of conditional distribution for X given $Y = y$ has conditional p.d.f. for each $y > 0$:

$$g_1(x|y) = \begin{cases} ye^{-xy} & \text{for } x \ge 0, \\ 0 & \text{otherwise.} \end{cases}$$

We shall assume that Y has a continuous distribution with p.d.f. $f_2(y) = e^{-y}$ for $y > 0$. Now we can construct the joint p.d.f. of X and Y using Theorem 3.6.2:

$$f(x, y) = g_1(x|y) f_2(y) = \begin{cases} ye^{-y(x+1)} & \text{for } x \ge 0, y > 0, \\ 0 & \text{otherwise.} \end{cases}$$

◀

Example 3.6.9 Defective Parts. Let X be the number of defective parts in a sample of size n, and let P be the proportion of defectives among all parts, as in Example 3.6.7. The joint p.f./p.d.f of X and $P = p$ was calculated there as

$$f(x, p) = g_1(x|p) f_2(p) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \text{for } x = 0, \ldots, n \text{ and } 0 \le p \le 1.$$

We could now compute the conditional p.d.f. of P given $X = x$ by first finding the marginal p.f. of X:

$$f_1(x) = \int_0^1 \binom{n}{x} p^x (1-p)^{n-x} dp, \tag{3.6.9}$$

The conditional p.d.f. of P given $X = x$ is then

$$g_2(p|x) = \frac{f(x, p)}{f_1(x)} = \frac{p^x (1-p)^{n-x}}{\int_0^1 q^x (1-q)^{n-x} dq}, \quad \text{for } 0 < p < 1. \tag{3.6.10}$$

The integral in the denominator of Eq. (3.6.10) can be tedious to calculate, but it can be found. For example, if $n = 2$ and $x = 1$, we get

$$\int_0^1 q(1-q) dq = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

In this case, $g_2(p|1) = 6p(1-p)$ for $0 \le p \le 1$. ◀

Bayes' Theorem and the Law of Total Probability for Random Variables The calculation done in Eq. (3.6.9) is an example of the generalization of the law of total probability to random variables. Also, the calculation in Eq. (3.6.10) is an example of the generalization of Bayes' theorem to random variables. The proofs of these results are straightforward and not given here.

Theorem 3.6.3 Law of Total Probability for Random Variables. If $f_2(y)$ is the marginal p.f. or p.d.f. of a random variable Y and $g_1(x|y)$ is the conditional p.f. or p.d.f. of X given $Y = y$, then the marginal p.f. or p.d.f. of X is

$$f_1(x) = \sum_y g_1(x|y) f_2(y), \tag{3.6.11}$$

if Y is discrete. If Y is continuous, the marginal p.f. or p.d.f. of X is

$$f_1(x) = \int_{-\infty}^{\infty} g_1(x|y) f_2(y)\, dy. \tag{3.6.12}$$

■

There are versions of Eqs. (3.6.11) and (3.6.12) with x and y switched and the subscripts 1 and 2 switched. These versions would be used if the joint distribution of X and Y were constructed from the conditional distribution of Y given X and the marginal distribution of X.

Theorem 3.6.4 Bayes' Theorem for Random Variables. If $f_2(y)$ is the marginal p.f. or p.d.f. of a random variable Y and $g_1(x|y)$ is the conditional p.f. or p.d.f. of X given $Y = y$, then the conditional p.f. or p.d.f. of Y given $X = x$ is

$$g_2(y|x) = \frac{g_1(x|y) f_2(y)}{f_1(x)}, \tag{3.6.13}$$

where $f_1(x)$ is obtained from Eq. (3.6.11) or (3.6.12). Similarly, the conditional p.f. or p.d.f. of X given $Y = y$ is

$$g_1(x|y) = \frac{g_2(y|x) f_1(x)}{f_2(y)}, \tag{3.6.14}$$

where $f_2(y)$ is obtained from Eq. (3.6.11) or (3.6.12) with x and y switched and with the subscripts 1 and 2 switched. ■

Example 3.6.10

Choosing Points from Uniform Distributions. Suppose that a point X is chosen from the uniform distribution on the interval [0, 1], and that after the value $X = x$ has been observed $(0 < x < 1)$, a point Y is then chosen from the uniform distribution on the interval $[x, 1]$. We shall derive the marginal p.d.f. of Y.

Since X has a uniform distribution, the marginal p.d.f. of X is as follows:

$$f_1(x) = \begin{cases} 1 & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, for each value $X = x$ $(0 < x < 1)$, the conditional distribution of Y is the uniform distribution on the interval $[x, 1]$. Since the length of this interval is $1 - x$, the conditional p.d.f. of Y given that $X = x$ will be

$$g_2(y|x) = \begin{cases} \dfrac{1}{1-x} & \text{for } x < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

It follows from Eq. (3.6.8) that the joint p.d.f. of X and Y will be

$$f(x, y) = \begin{cases} \dfrac{1}{1-x} & \text{for } 0 < x < y < 1, \\ 0 & \text{otherwise.} \end{cases} \tag{3.6.15}$$

Thus, for $0 < y < 1$, the value of the marginal p.d.f. $f_2(y)$ of Y will be

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y)\, dx = \int_0^y \frac{1}{1-x}\, dx = -\log(1 - y). \tag{3.6.16}$$

Furthermore, since Y cannot be outside the interval $0 < y < 1$, then $f_2(y) = 0$ for $y \le 0$ or $y \ge 1$. This marginal p.d.f. f_2 is sketched in Fig. 3.21. It is interesting to note that in this example the function f_2 is unbounded.

We can also find the conditional p.d.f. of X given $Y = y$ by applying Bayes' theorem (3.6.14). The product of $g_2(y|x)$ and $f_1(x)$ was already calculated in Eq. (3.6.15).

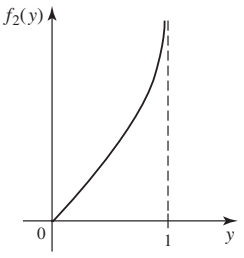


Figure 3.21 The marginal p.d.f. of Y in Example 3.6.10.

The ratio of this product to $f_2(y)$ from Eq. (3.6.16) is

$$g_1(x|y) = \begin{cases} \dfrac{-1}{(1-x)\log(1-y)} & \text{for } 0 < x < y, \\ 0 & \text{otherwise.} \end{cases}$$ ◀

Theorem 3.6.5 Independent Random Variables. Suppose that X and Y are two random variables having a joint p.f., p.d.f., or p.f./p.d.f. f. Then X and Y are independent if and only if for every value of y such that $f_2(y) > 0$ and every value of x,

$$g_1(x|y) = f_1(x). \tag{3.6.17}$$

Proof Theorem 3.5.4 says that X and Y are independent if and only if $f(x, y)$ can be factored in the following form for $-\infty < x < \infty$ and $-\infty < y < \infty$:

$$f(x, y) = f_1(x) f_2(y),$$

which holds if and only if, for all x and all y such that $f_2(y) > 0$,

$$f_1(x) = \frac{f(x, y)}{f_2(y)}. \tag{3.6.18}$$

But the right side of Eq. (3.6.18) is the formula for $g_1(x|y)$. Hence, X and Y are independent if and only if Eq. (3.6.17) holds for all x and all y such that $f_2(y) > 0$. ■

Theorem 3.6.5 says that X and Y are independent if and only if the conditional p.f. or p.d.f. of X given $Y = y$ is the same as the marginal p.f. or p.d.f. of X for all y such that $f_2(y) > 0$. Because $g_1(x|y)$ is arbitrary when $f_2(y) = 0$, we cannot expect Eq. (3.6.17) to hold in that case.

Similarly, it follows from Eq. (3.6.8) that X and Y are independent if and only if

$$g_2(y|x) = f_2(y), \tag{3.6.19}$$

for every value of x such that $f_1(x) > 0$. Theorem 3.6.5 and Eq. (3.6.19) give the mathematical justification for the meaning of independence that we presented on page 136.

Note: Conditional Distributions Behave Just Like Distributions. As we noted on page 59, conditional probabilities behave just like probabilities. Since distributions are just collections of probabilities, it follows that conditional distributions behave just like distributions. For example, to compute the conditional probability that a discrete random variable X is in some interval $[a, b]$ given $Y = y$, we must add $g_1(x|y)$ for all values of x in the interval. Also, theorems that we have proven or shall prove about distributions will have versions conditional on additional random variables. We shall postpone examples of such theorems until Sec. 3.7 because they rely on joint distributions of more than two random variables.

Summary

The conditional distribution of one random variable X given an observed value y of another random variable Y is the distribution we would use for X if we were to learn that $Y = y$. When dealing with the conditional distribution of X given $Y = y$, it is safe to behave as if Y were the constant y. If X and Y have joint p.f., p.d.f., or p.f./p.d.f. $f(x, y)$, then the conditional p.f. or p.d.f. of X given $Y = y$ is $g_1(x|y) =$

$f(x, y)/f_2(y)$, where f_2 is the marginal p.f. or p.d.f. of Y. When it is convenient to specify a conditional distribution directly, the joint distribution can be constructed from the conditional together with the other marginal. For example,

$$f(x, y) = g_1(x|y) f_2(y) = f_1(x) g_2(y|x).$$

In this case, we have versions of the law of total probability and Bayes' theorem for random variables that allow us to calculate the other marginal and conditional.

Two random variables X and Y are independent if and only if the conditional p.f. or p.d.f. of X given $Y = y$ is the same as the marginal p.f. or p.d.f. of X for all y such that $f_2(y) > 0$. Equivalently, X and Y are independent if and only if the conditional p.f. of p.d.f. of Y given $X = x$ is the same as the marginal p.f. or p.d.f. of Y for all x such that $f_1(x) > 0$.

Exercises

1. Suppose that two random variables X and Y have the joint p.d.f. in Example 3.5.10 on page 139. Compute the conditional p.d.f. of X given $Y = y$ for each y.

2. Each student in a certain high school was classified according to her year in school (freshman, sophomore, junior, or senior) and according to the number of times that she had visited a certain museum (never, once, or more than once). The proportions of students in the various classifications are given in the following table:

	Never	Once	More than once
Freshmen	0.08	0.10	0.04
Sophomores	0.04	0.10	0.04
Juniors	0.04	0.20	0.09
Seniors	0.02	0.15	0.10

a. If a student selected at random from the high school is a junior, what is the probability that she has never visited the museum?

b. If a student selected at random from the high school has visited the museum three times, what is the probability that she is a senior?

3. Suppose that a point (X, Y) is chosen at random from the disk S defined as follows:

$$S = \{(x, y) : (x - 1)^2 + (y + 2)^2 \le 9\}.$$

Determine **(a)** the conditional p.d.f. of Y for every given value of X, and **(b)** $\Pr(Y > 0|X = 2)$.

4. Suppose that the joint p.d.f. of two random variables X and Y is as follows:

$$f(x, y) = \begin{cases} c(x + y^2) & \text{for } 0 \le x \le 1 \text{ and } 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the conditional p.d.f. of X for every given value of Y, and **(b)** $\Pr(X < \frac{1}{2}|Y = \frac{1}{2})$.

5. Suppose that the joint p.d.f. of two points X and Y chosen by the process described in Example 3.6.10 is as given by Eq. (3.6.15). Determine **(a)** the conditional p.d.f. of X for every given value of Y, and **(b)** $\Pr\left(X > \frac{1}{2} \middle| Y = \frac{3}{4}\right)$.

6. Suppose that the joint p.d.f. of two random variables X and Y is as follows:

$$f(x, y) = \begin{cases} c \sin x & \text{for } 0 \le x \le \pi/2 \text{ and } 0 \le y \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the conditional p.d.f. of Y for every given value of X, and **(b)** $\Pr(1 < Y < 2|X = 0.73)$.

7. Suppose that the joint p.d.f. of two random variables X and Y is as follows:

$$f(x, y) = \begin{cases} \frac{3}{16}(4 - 2x - y) & \text{for } x > 0,\ y > 0, \\ & \quad \text{and } 2x + y < 4, \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the conditional p.d.f. of Y for every given value of X, and **(b)** $\Pr(Y \ge 2|X = 0.5)$.

8. Suppose that a person's score X on a mathematics aptitude test is a number between 0 and 1, and that his score Y on a music aptitude test is also a number between 0 and 1. Suppose further that in the population of all college students in the United States, the scores X and Y are distributed according to the following joint p.d.f.:

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & \text{for } 0 \le x \le 1 \text{ and } 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

a. What proportion of college students obtain a score greater than 0.8 on the mathematics test?

b. If a student's score on the music test is 0.3, what is the probability that his score on the mathematics test will be greater than 0.8?

c. If a student's score on the mathematics test is 0.3, what is the probability that his score on the music test will be greater than 0.8?

9. Suppose that either of two instruments might be used for making a certain measurement. Instrument 1 yields a measurement whose p.d.f. h_1 is

$$h_1(x) = \begin{cases} 2x & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Instrument 2 yields a measurement whose p.d.f. h_2 is

$$h_2(x) = \begin{cases} 3x^2 & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that one of the two instruments is chosen at random and a measurement X is made with it.

a. Determine the marginal p.d.f. of X.

b. If the value of the measurement is $X = 1/4$, what is the probability that instrument 1 was used?

10. In a large collection of coins, the probability X that a head will be obtained when a coin is tossed varies from one coin to another, and the distribution of X in the collection is specified by the following p.d.f.:

$$f_1(x) = \begin{cases} 6x(1-x) & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that a coin is selected at random from the collection and tossed once, and that a head is obtained. Determine the conditional p.d.f. of X for this coin.

11. The definition of the conditional p.d.f. of X given $Y = y$ is arbitrary if $f_2(y) = 0$. The reason that this causes no serious problem is that it is highly unlikely that we will observe Y close to a value y_0 such that $f_2(y_0) = 0$. To be more precise, let $f_2(y_0) = 0$, and let $A_0 = [y_0 - \epsilon, y_0 + \epsilon]$. Also, let y_1 be such that $f_2(y_1) > 0$, and let $A_1 = [y_1 - \epsilon, y_1 + \epsilon]$. Assume that f_2 is continuous at both y_0 and y_1. Show that

$$\lim_{\epsilon \to 0} \frac{\Pr(Y \in A_0)}{\Pr(Y \in A_1)} = 0.$$

That is, the probability that Y is close to y_0 is much smaller than the probability that Y is close to y_1.

12. Let Y be the rate (calls per hour) at which calls arrive at a switchboard. Let X be the number of calls during a two-hour period. Suppose that the marginal p.d.f. of Y is

$$f_2(y) = \begin{cases} e^{-y} & \text{if } y > 0, \\ 0 & \text{otherwise,} \end{cases}$$

and that the conditional p.f. of X given $Y = y$ is

$$g_1(x|y) = \begin{cases} \dfrac{(2y)^x}{x!} e^{-2y} & \text{if } x = 0, 1, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

a. Find the marginal p.f. of X. (You may use the formula $\int_0^\infty y^k e^{-y}\, dy = k!$.)

b. Find the conditional p.d.f. $g_2(y|0)$ of Y given $X = 0$.

c. Find the conditional p.d.f. $g_2(y|1)$ of Y given $X = 1$.

d. For what values of y is $g_2(y|1) > g_2(y|0)$? Does this agree with the intuition that the more calls you see, the higher you should think the rate is?

13. Start with the joint distribution of treatment group and response in Table 3.6 on page 138. For each treatment group, compute the conditional distribution of response given the treatment group. Do they appear to be very similar or quite different?

3.7 Multivariate Distributions

In this section, we shall extend the results that were developed in Sections 3.4, 3.5, and 3.6 for two random variables X and Y to an arbitrary finite number n of random variables $X_1, \ldots, X_n$. In general, the joint distribution of more than two random variables is called a multivariate distribution. *The theory of statistical inference (the subject of the part of this book beginning with Chapter 7) relies on mathematical models for observable data in which each observation is a random variable. For this reason, multivariate distributions arise naturally in the mathematical models for data. The most commonly used model will be one in which the individual data random variables are conditionally independent given one or two other random variables.*